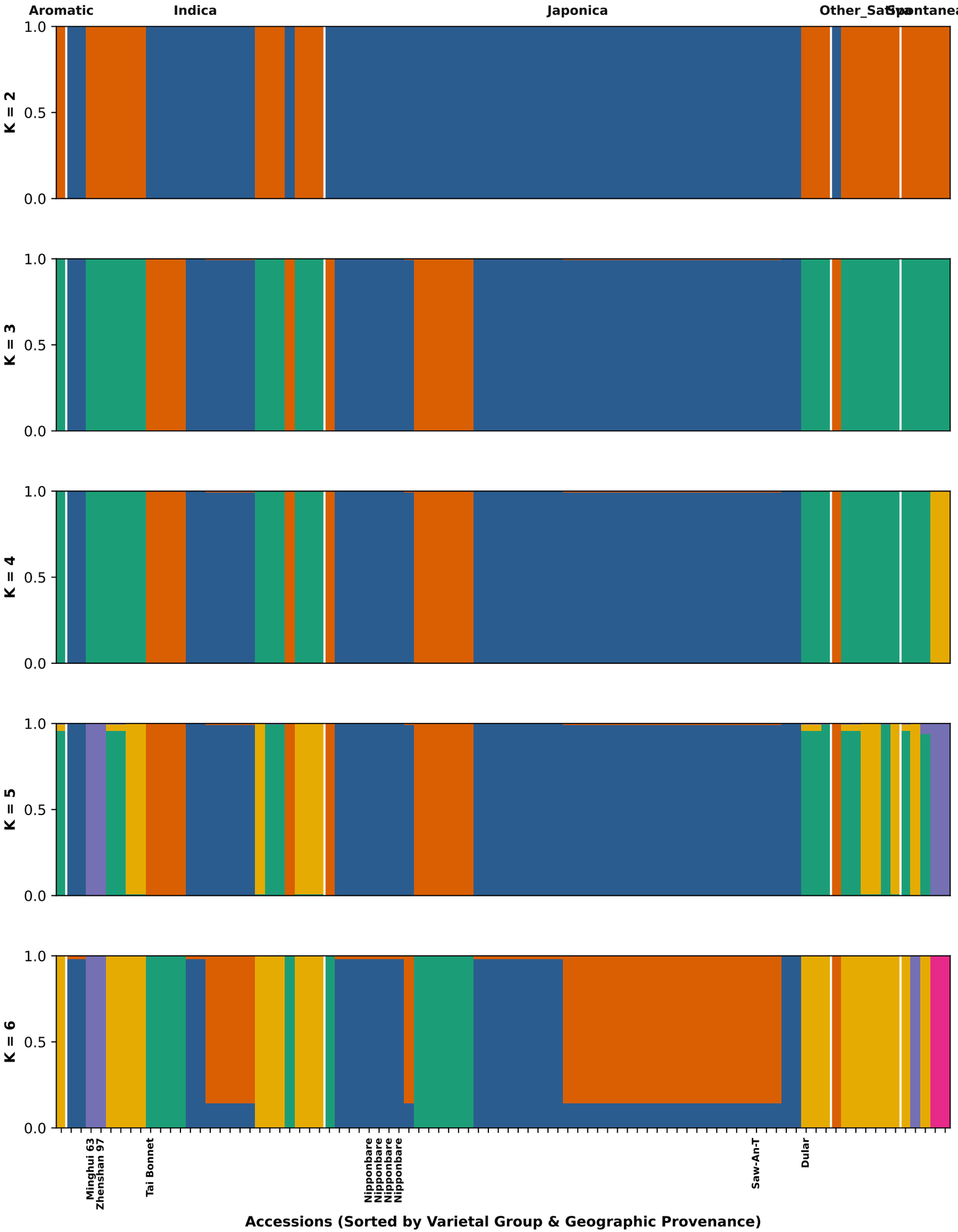
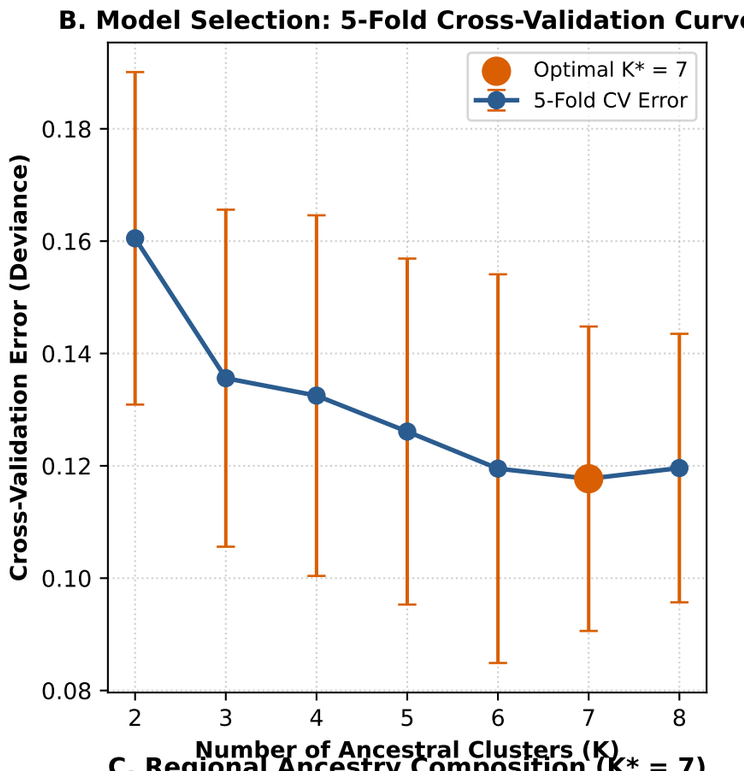
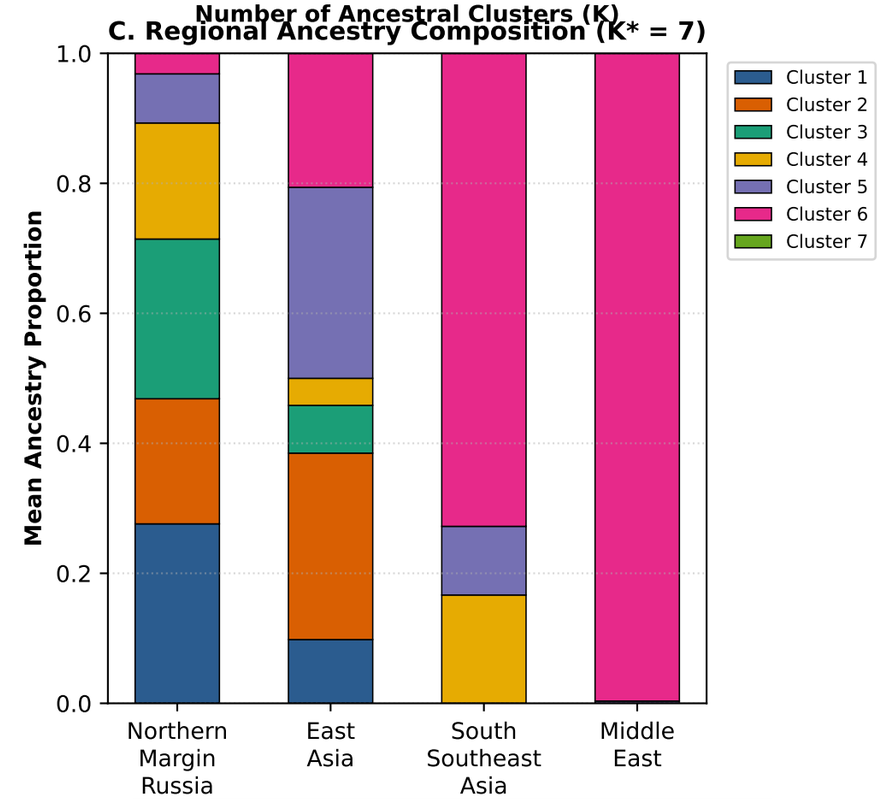






Admixture Optimization Invariants:
- Evaluated K = 2..8 over 20 independent tag SNPs.
- Optimal complexity: K* = 7 (CV deviance = 0.1177).
- Cluster 1 (#2b5c8f): Canonical Japonica maternal lineage.
- Cluster 2 (#d95f02): Canonical Lowland Indica lineage.
- Cluster 3 (#1b9e77): Northern margin Russian sub-lineage.
- Cluster 4 (#e6ab02): Ancestral wild Spontanea alleles.
- Cluster 5 (#7570b3): Aus / Aromatic early divergence.